Case Study 04: International Travel Expense & Currency Converter

1. Problem Statement

Develop a Python program to convert, track, and analyze international travel expenses incurred in foreign currency relative to an allocated budget in Indian Rupees (INR). The program reads the total budget in Rupees and the applicable exchange rate, captures expenditures across predefined categories ("Hotel", "Food", and "Transport"), applies a mandatory 2% currency conversion processing fee, calculates the overall cost, detects the largest expenditure category, and determines whether the traveler stayed within budget.

2. Objectives

- To read floating-point input values for budget in INR, currency exchange rate, and category-wise foreign expenses.
- To iterate through a list of expense categories and apply currency conversion formulas with a 2% transactional surcharge.
- To compute aggregate expenditure using generator expressions within the built-in sum() function.

- To determine the largest expense using max() with a lambda key selector and evaluate budget compliance using a ternary conditional operator.

3. Assumptions

- The conversion formula incorporates a flat 2% surcharge: converted $=$ amount $\times$ rate $\times$ 1.02.

- Expenses are recorded across three standard categories: Hotel, Food, and Transport.

- User inputs for budget, conversion rate, and expenses are positive numerical values.

- Currency figures are formatted and rounded to two decimal places.

4. Algorithm

- **Step 1:** Start the program.

- **Step 2:** Prompt the user to input budget in Rupees and rate (Rupees per unit of foreign currency) as floating-point numbers.

- **Step 3:** Initialize a list categories = ["Hotel", "Food", "Transport"] and an empty list expenses = [].

- **Step 4:** For each category in categories:

    - Read the foreign currency amount as amount.

    - Compute converted INR amount with 2% fee: converted = amount * rate * 1.02.

    - Append (category, converted) to expenses.

- **Step 5:** Compute aggregate expenses: total = sum(item[1] for item in expenses).

- **Step 6:** Identify the single largest category of expense: largest = max(expenses, key=lambda item: item[1]).

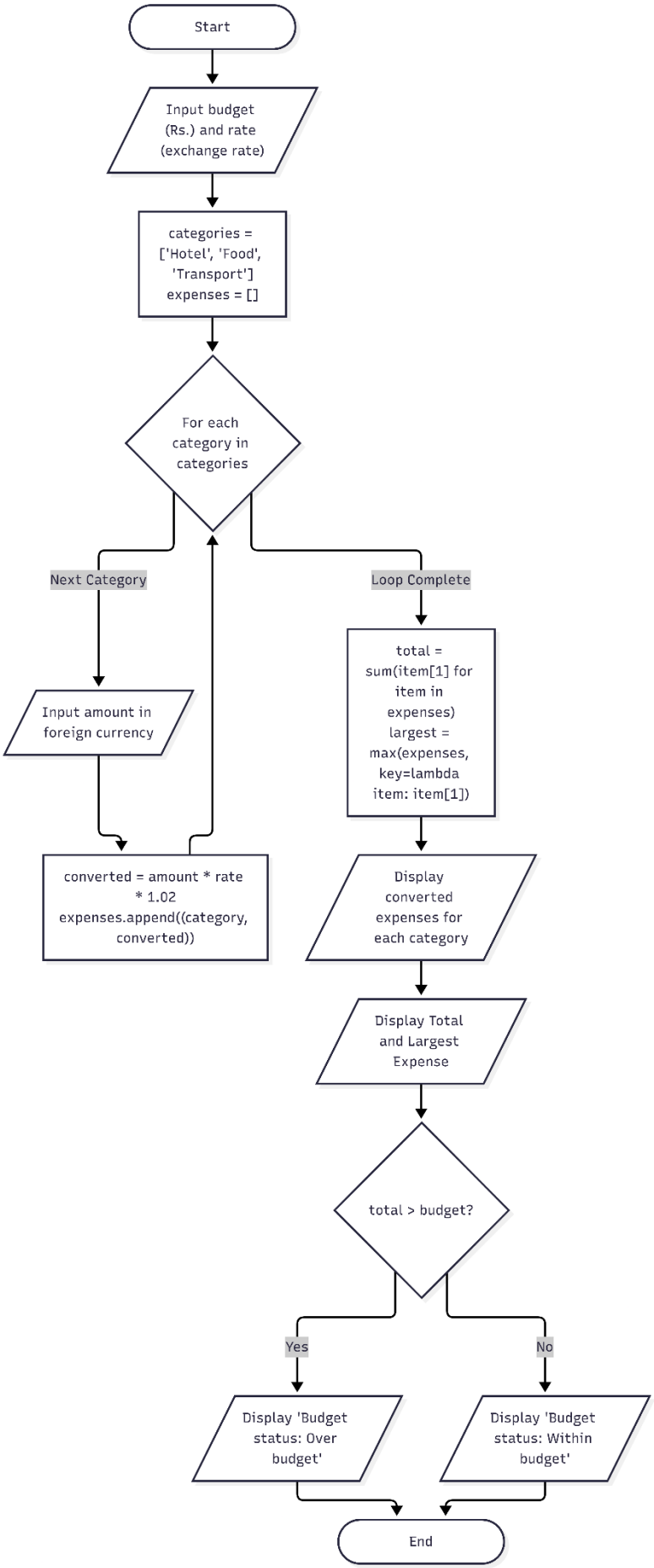
- **Step 7:** Display all converted category expenses rounded to 2 decimal places.

- **Step 8:** Display the total expenditure and the category name corresponding to largest[0].

- **Step 9:** Check if total > budget. If true, print "Budget status: Over budget"; otherwise, print "Budget status: Within budget".

- **Step 10:** Stop the program.

5. Flowchart Logic / Diagram


Start
Input budget (Rs.) and rate (exchange rate)
categories = ['Hotel', 'Food', 'Transport'] expenses = []
For each category in categories
Next Category
Input amount in foreign currency
converted = amount * rate * 1.02 expenses.append((category, converted))
Loop Complete
total = sum(item[1] for item in expenses) largest = max(expenses, key=lambda item: item[1])
Display converted expenses for each category
Display Total and Largest Expense
total > budget?
Yes
Display 'Budget status: Over budget'
No
Display 'Budget status: Within budget'
End

6. Pseudocode

```
Plaintext
BEGIN
  PRINT "Travel budget in Rs.: "
  READ budget
  PRINT "Rupees per 1 unit of foreign currency: "
  READ rate

  SET categories = ["Hotel", "Food", "Transport"]
  SET expenses = []

  FOR EACH category IN categories DO
    PRINT category + " expense in foreign currency: "
    READ amount
    SET converted = amount * rate * 1.02
    APPEND (category, converted) TO expenses
  END FOR

  SET total = SUM(amount FOR (_, amount) IN expenses)
  SET largest = MAXIMUM(expenses, BY amount)

  PRINT "Converted expenses including 2% fee:"
  FOR EACH (category, amount) IN expenses DO
    PRINT category, ": Rs.", ROUND(amount, 2)
```

```
  END FOR

  PRINT "Total: Rs.", ROUND(total, 2)
  PRINT "Largest expense:", largest[0]

  IF total > budget THEN
    PRINT "Budget status: Over budget"
  ELSE
    PRINT "Budget status: Within budget"
  END IF
END
```

7. Python Source Code

```
Python
# International Travel Expense & Currency Converter

# Read baseline financial parameters
budget = float(input("Travel budget in Rs.: "))
rate = float(input("Rupees per 1 unit of foreign currency: "))

# Define tracking categories and empty expense container
categories = ["Hotel", "Food", "Transport"]
expenses = []
```

```
# Collect, convert, and store expenses including a 2% conversion charge
for category in categories:
    amount = float(input(category + " expense in foreign currency: "))
    converted = amount * rate * 1.02
    expenses.append((category, converted))

# Compute total and identify maximum expense category
total = sum(item[1] for item in expenses)
largest = max(expenses, key=lambda item: item[1])

# Output itemized results and budget verification
print("Converted expenses including 2% fee:")
for category, amount in expenses:
    print(category, ": Rs.", round(amount, 2))
print("Total: Rs.", round(total, 2))
print("Largest expense:", largest[0])
print("Budget status:", "Over budget" if total > budget else "Within budget")
```

8. Test Cases

Test Case ID	Input Given	Expected Output	Actual Output	Status
TC_01 (Within Budget)	Budget = 50000	Total: Rs. 36720.0	Converted expenses including 2% fee:	Pass

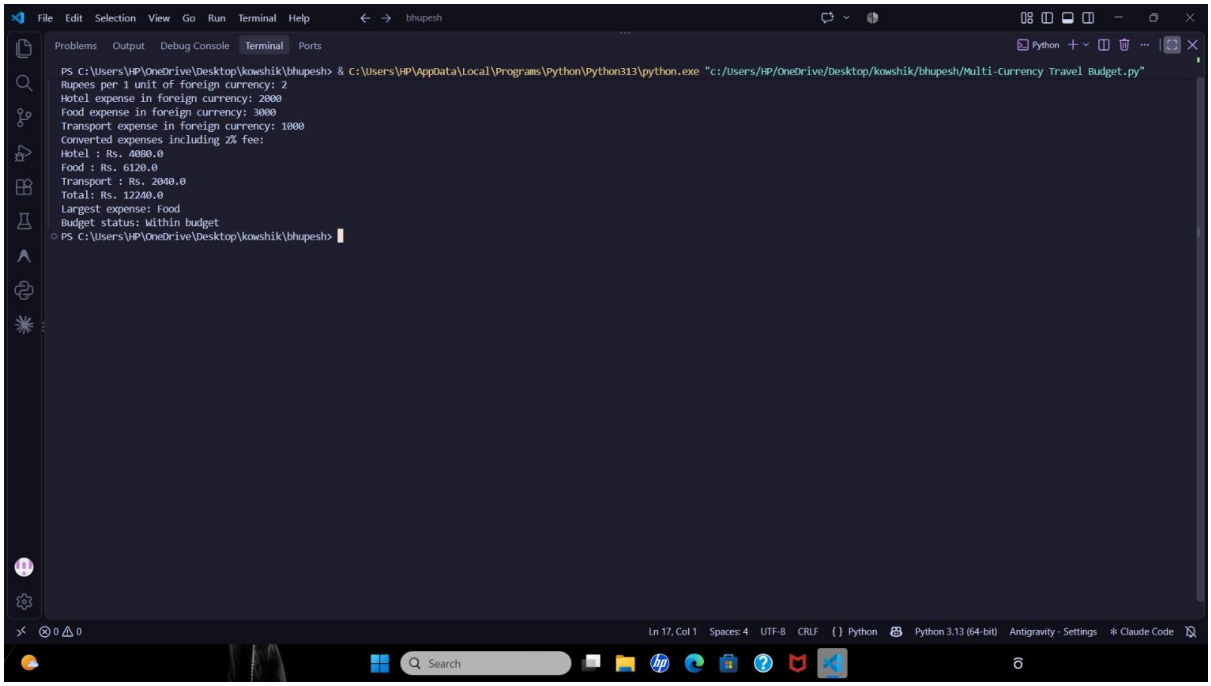
Test Case ID	Input Given	Expected Output	Actual Output	Status
	Rate = 80 Hotel = 200 Food = 100 Transport = 150	Largest expense: Hotel Budget status: Within budget	Hotel : Rs. 16320.0 Food : Rs. 8160.0 Transport : Rs. 12240.0 Total: Rs. 36720.0 Largest expense: Hotel Budget status: Within budget	
TC_02 (Over Budget)	Budget = 20000 Rate = 90	Total: Rs. 27540.0	Converted expenses including 2% fee:	Pass

Test Case ID	Input Given	Expected Output	Actual Output	Status
	Hotel = 150 Food = 100 Transport = 50	Largest expense: Hotel Budget status: Over budget	Hotel : Rs. 13770.0 Food : Rs. 9180.0 Transport : Rs. 4590.0 Total: Rs. 27540.0 Largest expense: Hotel Budget status: Over budget	
TC_03 (Transport Highest)	Budget = 60000 Rate = 85	Total: Rs. 56355.0 Largest expense: Transport	Converted expenses including 2% fee: Hotel : Rs. 8670.0	Pass

Test Case ID	Input Given	Expected Output	Actual Output	Status
	Hotel = 100 Food = 150 Transport = 400	Budget status: Within budget	Food : Rs. 13005.0 Transport : Rs. 34680.0 Total: Rs. 56355.0 Largest expense: Transport Budget status: Within budget	
TC_04 (Exact Budget Boundary)	Budget = 83300 Rate = 81.6667	Total: Rs. 83300.03 Budget status: Over budget	Converted expenses including 2% fee: Hotel : Rs. 41650.02	Pass

Test Case ID	Input Given	Expected Output	Actual Output	Status
	Hotel = 500 Food = 250 Transport = 250		Food : Rs. 20825.01 Transport : Rs. 20825.01 Total: Rs. 83300.03 Largest expense: Hotel Budget status: Over budget	

9. Output Screenshots



10. Conclusion

The International Travel Expense & Currency Converter program was successfully executed and evaluated. The script demonstrates practical use of list iteration, dynamic tuple packing, inline fee computation, generator-based aggregation via sum(), and ternary conditionals for status reporting.